

			
	Ma'aden Phosphate 3 Phase I		
Project Phase :	Stage 4 - EPCM	Ma'aden Project N° :	18-PME-PH3-PH1-001
Main Title:	SPECIFICATION-ANCHOR BOLTS	Document N°:	MD-520-1000-EG-CV-SPC-0004
		Revision:	C01

ISSUED FOR CONSTRUCTION

MA'ADEN

Phosphate 3 Phase I – Stage 4

Specification Anchor Bolts

		Ma'aden Phosphate 3 – Phase 1	
NAME	TITLE	SIGN	DATE
REVIEW GRADE			
Result	Code	Description	
☐	A	Accepted	
☐	C	Proceed with exceptions to next Submission and Revision	
☐	R	Do Not Proceed - Revise as noted and Resubmit	
Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due and proper performance of the Works in accordance with the Contract/Purchase Order.			
PROJECT MANAGER SIGN-OFF:			
NAME		DATE	

Rev.	Issue Date	Revision Description	Prepared by	Checked by	Discipline Approved	Project Approved by
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1. INTRODUCTION

1.1 Project Introduction

The Saudi Arabian Mining Company (Ma'aden) is proceeding with the development of two strategic projects:

1-Ras Al Kair project: Two Diammonium phosphate (DAP) plants (phase 1) with all associated facilities / utilities and storage in the area adjacent to the Eastern side of Ma'aden Phosphate Company (MPC) at Ras Al-Khair

2- Waad Al Shamal project: the proposed site includes a mine and related infrastructure works as well as chemical plants associated with the processing of phosphate in the production of fertilizers including related infrastructures such as roads and railways etc. located within 5km from the mine area. The project comprises also a dam for flood protection and a gypsum storage.

1.2 Project Description

1.2.1 Ras Al Khair project (RAK)

The scope of service for RAK describes without limitation the design, procurement and building of an overall chemical complex for the production of Fertilizers.

The scope of work covers the following plants and facilities

- Granulation Plant: 2 Trains of 1 million MTPY (DAP Equivalent) per train to produce different fertilizers grades.
- Material Handling and Storage Halls.
- Connections at the battery limits
- All other utilities, equipment and material required to ensure safety, operability, maintainability, process performances and reliability for all the plants life cycle.

1.2.2 Wa'ad Al Shamal project (WAS)

The scope of service for WAS concerns the design, procurement and building of an overall chemical complex for the whole Phosphoric acid production value chain, starting from the mining crushing area to the delivery of MGA at the battery limits.

The scope covers the following plants and facilities

- Crushing area, including its electrical substation
- Beneficiation Plant of 2.05 Million MT / year, Technology provided by FLS
- Phosphate tailing facility
- SAP of 1.75 Million MT / year, licensed by Metso: Outotec
- Power Plant of approximately 65 MW
- PAP of 0.5 Million MTP2O5 / year, licensed by Prayon

			
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- Gypsum Storage facility
- Water Treatment area
- Connections at the battery limits
- All other utilities, equipment and material required to ensure safety, operability, maintainability, process performances and reliability for all the plants life cycle.

1.3 Scope

This document provides the details and requirements for fabrication and installation of all Major anchor bolts (e.g. for building columns, trestle columns, tank columns and pipe racks) shall be cast into concrete. Anchor bolt assemblies that include bolt threaded at the two end, nuts, washer, and anchor plates and/or sleeves if required. Three standard types are shown for each diameter anchor bolt in metric units in the anchor bolt standard drawing (MD-520-1000-EG-CV-STD-0005).

Any special type of anchor bolt designed and supply by Mech/ Elect vendor are excluded from this specification.

This document is intended to be used in conjunction with the applicable documents listed herein for use in Ma'aden Phosphate 3 Phase I.

1.4 Purpose

The purpose of this specification is to provide details and requirements for anchor bolt fabricators and installers.

2. KEY TERMS AND DEFINITIONS

ACI	American Concrete Institute
ASTM	American Society for Testing and Materials
SBC	Saudi Building Code
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
EPCM Contractor	Worley
Contractor	An entity, engaged or approved by the employer, entrusted with a contract to perform work or cause work to be performed.
Manufacturer or Vendor	VENDOR responsible for design, sourcing of materials, manufacturing, assembling, testing and supply of equipment

			
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Work	The entire work and services and all things which the contractor is or may be required to execute, supply and do under the contract having regard to the scope of project, or to cause to be executed, supplied or done, for the purpose of constructing and commissioning the plant
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3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the revision of the relevant standards or codes mentioned in structural design criteria of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 Local Codes and Standards

3.2.1.1 Saudi Building Code

SBC 303	Structural – Soil and Foundation
SBC 304	Structural – Concrete Structure
SBC 306	Structural – Steel Structures.

3.2.2 International Codes and Standards

3.2.2.1 American Society for Testing and Materials (ASTM)

ACI 318-11	Building Code Requirements for Reinforced Concrete and Commentary.
AISC 360 -10	Specification for Structural Steel Buildings

			
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ASTM A6/6M	Rolled Structural Steel Bars, Plate Shapes and Sheet Piling.
ASTM A36/A36M	Standard Specification for Carbon Structural Steel.
ASTM A153/A153M	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM A193/A193M	Standard Specification for Alloy Steel and Stainless Steel Bolting Material for High Temperature Service.
ASTM A194/A194M	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High Temperature Service.
ASTM A307	Specification for Carbon Steel Bolts and Studs, 60000 psi Tensile Strength.
ASTM F436	Standard Specification for Hardened Steel Washers.
ASTM A563	Standard Specification for Carbon and Alloy Steel Nuts.
ASTM F1554	Standard Specification for Anchor Bolts, Steel, 36, 55, 105 ksi Yield Strength.
ASTM F2329	Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts and Special Threaded Fasteners.
API RP 686	Recommended Practice for Machinery Installation and Installation Design

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0001	Steelwork Fabrication.
MD-101-SMEM-EG-ST-SPC-0002	Steelwork Erection.
MD-101-SMEM-EG-CV-SPC-0002	Anchor Bolts
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

3.4 Project Standards

MD-520-1000-EG-ST-DST-0005	Structural Standards Anchor Bolt Details Drawing
MD-520-1000-EG-ST-SPC-0003	Specification-Grout
MD-520-1000-EG-CV-SPC-0002	Concrete Works specification.
MD-520-1000-EG-ST-SPC-0002	Painting Specifications.
MD-520-1000-EG-CV-CRT-0001	Civil and Structural Design Criteria

			
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4. MANUFACTURE

4.1 Material

Generally, anchor bolts shall conform to ASTM F1554 Section 6 and Table 3.

Nuts shall conform to ASTM A563, Grade A, heavy hex with UNC-2B threads. Washers shall be ASTM F436.

Anchor plate shall conform to ASTM A36.

4.2 Fabrication

Unless otherwise specified, all bolts (total bolt length), nuts and washers shall be hot-dip galvanized after fabrication in accordance with ASTM A153, Class C.

Excess galvanizing material shall be removed from the threaded portions of the bolts by the use of a centrifuge or by mechanical chasing of the threads.

The fit of nuts on threads of anchor bolts shall be verified before shipment.

4.3 Tolerances

The tolerances on overall lengths of any anchor bolt shall be +/- 5 mm, hooks shall have a tolerance +/- 6 mm on overall dimensions after bending.

The tolerance in setting out bolts in group with a specified template shall comply with ACI 318-11.

5. INSTALLATION

5.1 General

Anchor bolts details type A, B and N are shown on the Standard Drawing MD-520-1000-EG-ST-DST-0005.

Type A standard length anchor bolt assemblies shall consist of anchor bolts with tack-welded nut.

Type B extra-long length anchor bolt assemblies shall consist of anchor bolts with plate and tack-welded nut.

Type N non-standard length anchor bolt assemblies shall consist of anchor bolts with tack-welded nut.

Unless noted otherwise on the design drawings, anchor bolt sleeves for stationary equipment and structural base plates shall be filled with grout after the column or equipment is in place and the anchor bolts are aligned. Water or other loose particles shall not be allowed to collect in the sleeve before the sleeve is grouted. Grouting procedure shall conform to the Standard Specification MD-520-1000-EG-ST-SPC-0003, Grout specification.

			
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For machines and equipment anchor bolts, machine/equipment vendor shall specify anchor bolts size and spacing to the Employer for incorporation with the machine/equipment support design.

Unless noted otherwise on the design drawings, anchor bolts sleeves for machinery shall be filled with non-bonding moldable material before grouting.

Anchor bolt threads shall be covered with duct tape or other suitable means to keep them clean and to prevent any damage that might occur during the preparation of the foundation for grouting and the actual grouting of the foundation.

Unless a torque value is specified on design drawings or at the locations of slide plates, anchor bolts shall be tightened to a snug condition. At slide plate locations, nuts shall be backed off a half turn and a double nut shall be added.

All anchor bolts subjected to sulfurous gases, mists and fumes as well as high humidity shall be protected against corrosion by painting to the same paint system as structural steel after the structural steel is erected or the associated equipment is installed. The Contractor shall confirm surface preparation and primer type are acceptable for chosen paint manufacturer. All anchor bolts subjected to sulfurous gases, mists and fumes as well as high humidity shall be protected against corrosion with the same protective coating that applied to the structural steel, which is specified in the standard specification MD-520-1000-EG-ST-SPC-0002, Painting specification.

5.2 Sleeves

Anchor Bolt Sleeves shall be cylindrical type, high density polyethylene construction with 2.5 mm minimum wall thickness, or other material as shown on the Design Drawings. Sleeve diameter and length shall be as indicated on the Design Drawings.

Anchors should be installed with sleeves when small movement of the bolt is desired after the bolt is set in concrete. The two most common examples follow:

- When precise alignment of anchors is required during installation of structural columns or equipment. In this situation, the sleeve should be filled with grout after installation is complete.
- When anchors will be pre-tensioned to maintain the bolt under continuous tensile stresses during load reversal. Pre-tensioning requires the bolt surface to be free; therefore, the top of these sleeves should be sealed, or the sleeve should be filled with elastomeric material to prevent grout or water from filling the sleeve.

Refer to API RP 686 for use of sleeves with anchor bolts in machinery foundations chapter – 4 Section 3.6.

			
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6. QUALITY ASSURANCE

6.1 General

A Quality Assurance/Inspection Plan shall be provided, detailing procedures to be employed and the responsible personnel, to ensure that the requirements of this Specification and the applicable codes and standards are met as specified in MD-101-SMPM-PM-QA-PEI-0001, Ma'aden Contractor Quality Requirements Project Execution Instruction.

The Employer shall have the authority to reject materials, procedures or work which, in its opinion fail to meet the requirements of the Specification, the Contract, or Documents referred to in this Specification.

6.2 Inspection Requirement

The following inspections shall be carried out by the Contractor as a minimum.

6.2.1 Material

Inspection procedures shall be developed to ensure that materials meet the requirements of Steelwork Construction Specification, prior to installation.

6.2.2 Identification Requirement

Group of identical bolts shall be securely tied together for shipment and tagged according to concrete drawing number followed by schedule item number.

Tags shall consist of stamped aluminium or zinc strips securely wired to the bolts.

6.2.3 Bolt Protection

Visual inspection of surface treatment shall be carried out. Any defects of the bolts shall be reported and to be replaced. Installed anchor bolts shall be cover with a plastic cap. ASTM A780M practice can be used as per site condition to repair hot dip galvanized coating damaged by excessively rough handling during shipping or erection.

6.2.4 Field Welding

Welding inspection and testing frequency for field welding shall be the same as specified for shop welding in MD-520-1000-EG-ST-SPC-0001_Structural Steel Construction Specification.

The Contractor shall specify in the inspection testing plan the methodology to be employed in the sampling and selection of the welder and the welds to be inspected.

6.2.5 Bolting

All fabricated anchor bolts shall be tightened to the snug tight condition. All proprietary anchor bolts shall be tightened to the manufacturer's recommended torque unless noted otherwise on the design drawings. All fixings including nuts, bolts and washers shall be hot dip galvanized in accordance with ASTM A153M.